

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and Environmental Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and Environmental Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCEC604 - PROJECT MANAGEMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 In PERT analysis, the standard deviation of the overall project completion time can be estimated by
 - A) Equal to expected time
 - B) Square root of the sum of variances along the critical path
 - C) Difference between pessimistic and optimistic times
 - D) Sum of activity durations
 - 2 Which of the following options accurately describes how the value of money changes over time?
 - A) Money received in the future is worth more than the same amount received today
 - B) Money received today is worth more than the same amount received in the future
 - C) Money has the same value regardless of when it is received
 - D) Money never earns interest
 - 3 Which probability distribution is typically used to approximate the total project duration in PERT?
 - A) Beta distribution
 - B) Normal distribution
 - C) Poisson distribution
 - D) Uniform distribution
 - 4 What is the key purpose of using project management software?
 - A) Designing hardware circuits
 - B) Assigning tasks, allocating resources, and tracking project progress
 - C) Encrypting financial data only
 - D) Creating graphic animations

- 5 What is the probability that a project will be completed by its expected completion time
 - A) 0%
 - B) 50%
 - C) 100%
 - D) 25%
- 6 Computer-aided project management tools are mainly used to:
 - A) Design circuit diagrams
 - B) Plan, schedule, and monitor project activities
 - C) Create 3D models
 - D) Edit videos
- 7 Which probability distribution is normally used to model the time estimates of individual activities in PERT?
 - A) Normal distribution
 - B) Beta distribution
 - C) Gamma distribution
 - D) Exponential distribution
- 8 In project scheduling, what term describes the estimated amount of time needed to complete an activity?
 - A) Float
 - B) Duration
 - C) Slack
 - D) Lag
- 9 During the execution phase of a project, updating the CPM network is primarily done to:
 - A) Recalculate the total project budget only
 - B) Update activity durations and recalculate early/late times and critical paths
 - C) Increase activity floats deliberately
 - D) Remove critical path activities
- 10 When you crash a project to reduce its overall duration, which activities should be targeted first?
 - A) Activities with the shortest durations
 - B) Non-critical activities
 - C) Critical path activities with the lowest cost slope
 - D) Activities with the largest float
- 11 Which tool is most effective for illustrating the dependencies and sequencing of project activities?
 - A) Milestone chart
 - B) Bar (Gantt) chart without dependencies
 - C) Project network diagram (PDM)
 - D) Project budget table
- 12 A Gantt chart (bar chart) is mainly used to show:
 - A) Only the project milestones, ignoring task durations
 - B) Tasks and their durations along a timeline
 - C) The logical dependencies without any time scale
 - D) Only resource assignment details

Q2. Solve the following.

- A) Explain the differences between the Critical Path Method (CPM) and the Program Evaluation and Review Technique (PERT).

B) Describe the different phases of Project Management. 6

Q3. Solve the following.

A) Describe how updating calculations is done in a project network and why updating is important during project execution. 6

B) 6

Activity	Normal Time	Crash Time	Normal Cost	Crash Cost
1-2	5	3	5000	10000
1-3	2	1	1500	3000
2-4	5	3	2000	3000
2-5	7	3	5000	8000
3-5	6	3	8000	10000
4-6	8	4	5000	6000
5-6	9	5	15000	18000

Crash the network given above and determine the Project Duration (in Days) and Project Cost.

Q4. Solve Any Two of the following.

A) Discuss the central limit theorem in detail 6

B) Explain the Direct Cost and indirect cost for a construction project. 6

C) 6

Activity	Optimistic Time	Most Likely time	Pessimistic time
1-2	6	10	20
1-3	7	10	13
2-4	9	15	21
2-5	7	11	15
3-5	3	7	11
4-6	9	12	15
5-6	7	9	11

Draw the network and determine the Project Duration.

Find the Z value if the Project mentioned above has to be completed in 22 days

Q5. Solve Any Two of the following.

A) Write a short note on Demand and Supply. 6

B) What is the time value of money, and why is it important in financial decision-making? 6

C) 6

Activity (i-j)	1-2	2-3	2-4	3-5	3-6	4-5	4-7	5-8	6-8	7-8
Duration (t _{ij})	5	2	6	4	4	2	3	7	8	2

Draw the network and calculate TE and TL for all activities for the following.

Q6. Solve Any Two of the following.

A) A company manufactures a product with the following details: 6

Selling price per unit = ₹300

Variable cost per unit = ₹150

Total fixed costs = ₹4,50,000

a) Calculate the break-even point (BEP) in units.

b) Find the break-even sales revenue in rupees.

B) Discuss how ISO standards help improve project management practices in organisations. 6

- C) Explain the concept of the Juran Trilogy and its role in improving organisational quality. 6

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